Data sheet Cisco public

AI Solutions with Cisco Nexus Hyperfabric

Contents

Overview

Cisco Nexus One

Cisco Nexus® One represents a paradigm shift– an open, unified networking architecture that integrates silicon, systems, software, a single operating model, security and observability. Designed to adapt to customers' unique needs and stages of network transformation, Cisco Nexus One provides flexibility and interoperability across the stack, while maintaining operational consistency across the network. The philosophy is simple: "Open Choice. No Compromises." Whether you are building massive AI clusters or enterprise clouds, Cisco Nexus One provides the consistency and performance you need.

At the top of the Nexus One framework lies the operating model. This layer is designed to orchestrate across multiple sites and fabrics, streamlining how you scale and operate networks. The goal is to deliver a consistent experience, regardless of where your workloads live. The operating model offers an on-premesis approach with Cisco Nexus Dashboard and a cloud managed approach with Cisco Nexus Hyperfabric™.

Cisco Nexus Hyperfabric

Cisco Nexus Hyperfabric offers a flexible and scalable AI infrastructure solution designed to meet diverse customer needs. It provides two distinct deployment options that cater to different levels of integration and management preferences: one that enables connection to any AI server and storage platform through cloud-managed networking, and another that delivers a comprehensive full-stack infrastructure with pre-integrated compute, storage, and networking components. These options allow organizations to choose the approach that best fits their AI workload requirements and operational models.

- **Option 1:** Connect to any AI server and storage platform with cloud-managed networking

  This option provides a cloud-managed Hyperfabric network that enables flexible Ethernet fabric connectivity to any AI server platform and storage system. It offers pre-configured, cloud-managed Cisco® switches optimized for AI workloads, delivering high-performance, low-latency, and lossless backend networking. This approach supports diverse AI servers and storage platforms, allowing customers to bring their own servers, GPUs, storage, and software stacks while benefiting from simplified lifecycle management including design, order, deployment, monitoring, and upgrades. The cloud-managed controller provides unified visibility and control over the AI fabric, facilitating streamlined operations and scalability without requiring a full-stack infrastructure purchase.

- **Option 2:** Full-stack infrastructure option

  The full-stack option delivers a turnkey, vertically integrated AI infrastructure solution that includes Cisco UCS servers equipped with NVIDIA GPUs, partner storage (such as VAST Data), and Cisco Hyperfabric networking. This option is NVIDIA Enterprise Reference Architecture (ERA) compliant, providing a validated blueprint for scalable, high-performance AI infrastructure. It features pre-configured, cloud-managed networking combined with high-performance compute servers and optional high-throughput storage. The full-stack solution simplifies AI cluster provisioning and scaling by integrating compute, networking, and storage into a unified, cloud-managed platform with automated lifecycle management from design through deployment and operations.

Cloud-managed networking: connect to any AI server and storage platform

Obtain the benefits of a cloud-managed Hyperfabric network that offers flexible connectivity to any AI server and storage platform while delivering simplified operations and optimized high performance specifically designed for AI workloads.

Key components for the cloud-managed networking option (option 1) of Cisco Nexus Hyperfabric include:

- **Cloud-hosted Cisco Nexus Hyperfabric controller** that manages the network fabric configuration and complete lifecycle (design, ordering, deployment, management).
- **High-performance Ethernet fabric:** The fabric is designed for AI workloads with pods of plug-and-play leaf-spine fabrics, enabling flexible architectures that can connect to any AI server and storage platform.
- **Cisco Nexus Hyperfabric switches:** Specific models such as HF6100-64ED (800G spine), HF6100-32D (400G spine), and HF6100-60L4D (leaf) are used. These switches feature Cisco Silicon One® technology, support high port densities and are optimized for AI workloads with cloud-managed operation. Additionally, Cisco N9300 Series Switches are available and further enhance this high-performance, cloud-managed AI networking solution, ensuring simplified operations and optimized performance for AI workloads.
- **NVIDIA Spectrum4-based Cisco N9100 Series Switches with 800 GbE capacity:** This solution delivers a high-performance, NCP Reference Architecture (NCP RA)-compliant design tailored for next-generation secure AI factories, ensuring simplified operations and optimized performance for AI workloads.
- **Unified operations:** The cloud controller manages the AI fabric lifecycle, including design, deployment, monitoring, upgrades, and troubleshooting. It provides a unified operational experience with assertion-based event correlation and proactive alerting.

Cisco Nexus Hyperfabric's full-stack AI infrastructure option

The deployment model for the Cisco Nexus Hyperfabric's full-stack AI infrastructure option is turnkey, vertically integrated, and NVIDIA ERA-compliant, consisting of networking, servers, GPUs, and storage. It is cloud-managed to simplify the entire lifecycle, from design to operation, for demanding AI enterprise workloads. This option optimizes for time to first token for dedicated AI clusters from concept to production.

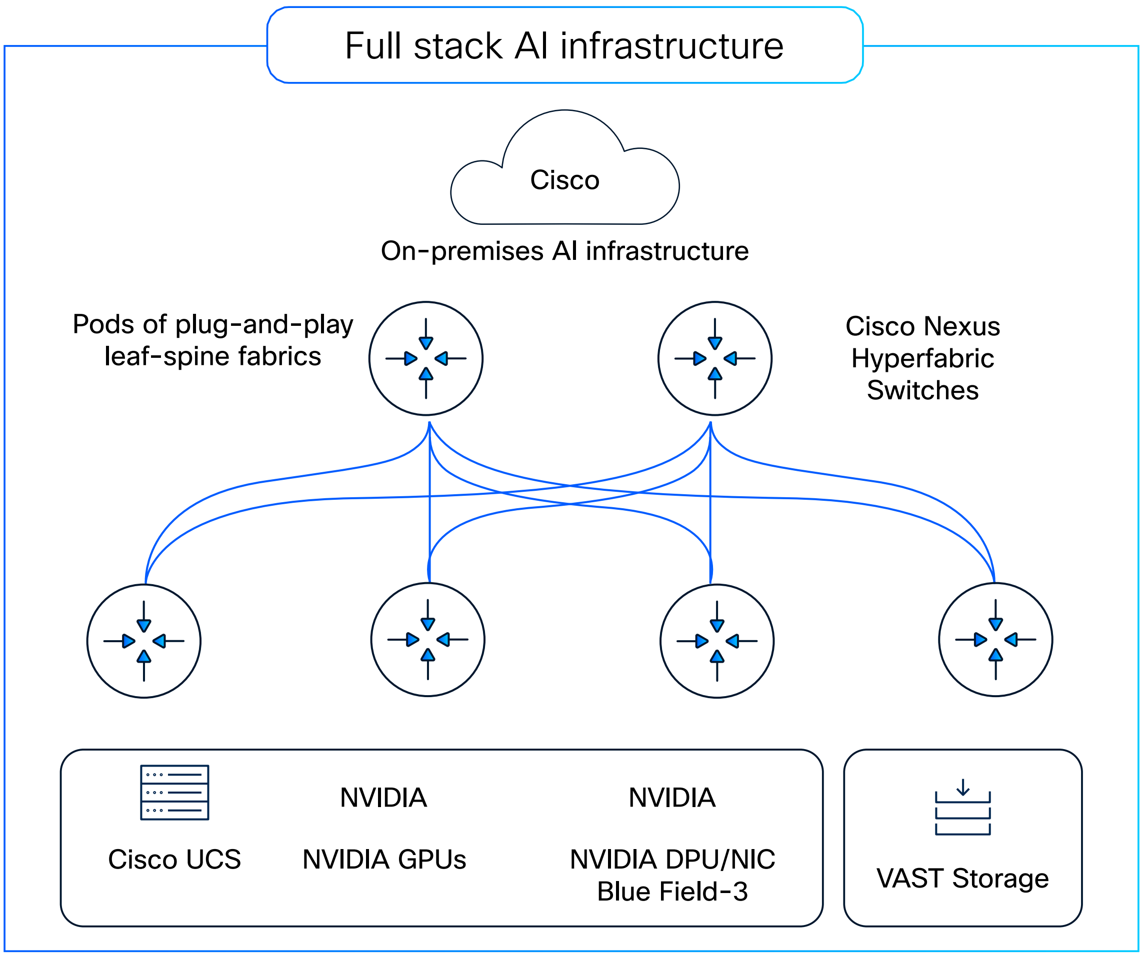

Figure 1. Cisco Nexus Hyperfabric's full-stack AI infrastructure option

Cisco Nexus Hyperfabric's full-stack AI infrastructure option is a turnkey, on-premises, AI-cluster option managed by a cloud-hosted controller, designed to simplify and accelerate AI deployments for enterprises and neocloud. It simplifies AI infrastructure deployment with a comprehensive, integrated, cloud-managed option compliant with NVIDIA Enterprise Reference Architecture (ERA). This cloud-managed vertical stack option consists of purpose-built hardware, software, a Cisco Nexus Hyperfabric cloud controller, day-2 operations, automation, and Cisco support that eliminates complexity. IT, application, and/or DevOps teams manage the full lifecycle of designing, ordering, deploying, validating, monitoring, and scaling fabrics without requiring deep networking or operational expertise.

It integrates Cisco's high-performance networking hardware, including Cisco Silicon One switches and Cisco UCS C885A M8 Rack Servers with NVIDIA HGX, with a cloud-managed operational model. This option offers preconfigured templates for AI clusters, enabling rapid design, ordering, deployment, and scaling of AI infrastructure with zero-touch provisioning and continuous monitoring. It is a unified stack that includes networking, compute, GPUs, AI software, and storage components, all managed through a cloud controller to provide full visibility and control.

Cisco Nexus Hyperfabric's full stack AI infrastructure option is built to deliver a simplified, scalable, and secure AI infrastructure that aligns with NVIDIA's Enterprise Reference Architecture, facilitating efficient AI workloads and innovation across industries.

Key components:

- Cloud-hosted Cisco Nexus Hyperfabric controller that manages network fabric configuration and lifecycle (design, ordering, deployment, management, and scaling of AI compute, GPUs, storage, and network infrastructure)
- AI cluster templates for small to extra-large deployments
- Cisco Nexus 6000 Series Switches with Silicon One technology, including HF6100 and the Cisco N9300 Series Switches
- Now featuring the NVIDIA Spectrum4-based Cisco N9100 Series Switches with 800 GbE capacity. This solution delivers a high-performance, NCP Reference Architecture (NCP RA)–compliant design tailored for next-generation secure AI factories, helping to ensure simplified operations and optimized performance for AI workloads
- Cisco UCS C885A M8 Rack servers equipped with NVIDIA HGX H200 GPUs and BlueField-3 DPUs and UCS 880 equipped with HGX B300 (coming soon)
- Optional VAST data-storage platform integrated with Cisco UCS servers
- Cisco optics and cables
- Optional AI software stack including NVIDIA AI Enterprise (NVAIE)
- Onsite installation services by Cisco Customer Experience (CX)

Performance and features:

- NVIDIA ERA–compliant for AI workloads, ensuring validated performance and scalability
- High-throughput fabric with 800 GB/s bandwidth and ultra-low latency
- Support for RDMA, NVMe Over Fabrics (NVMeoF), and advanced telemetry
- Cloud-managed lifecycle with zero-touch provisioning and continuous software and firmware updates
- End-to-end assertion-based monitoring from network fabric to compute NICs and storage
- Preconfigured leaf-and-spine pods for plug-and-play deployment

Scalability:

- Modular design supporting small, medium, and large AI clusters
- Pre-configured designs for the entire infrastructure component suite – frontend, backend, rails, storage, and management
- Independent scaling of network, compute, GPUs, and storage resources
- Fully integrated turnkey option with VAST storage or support for Bring Your Own (BYO) storage

Use cases and benefits:

- Large-scale enterprise AI model training, fine tuning, and inference
- Dedicated AI/ML clusters with optimized efficiency and low latency
- Faster time to value with replicable deployment models
- Validated turnkey AI stack through Cisco, NVIDIA, and storage vendor partnership, ensuring optimized hardware and software integration

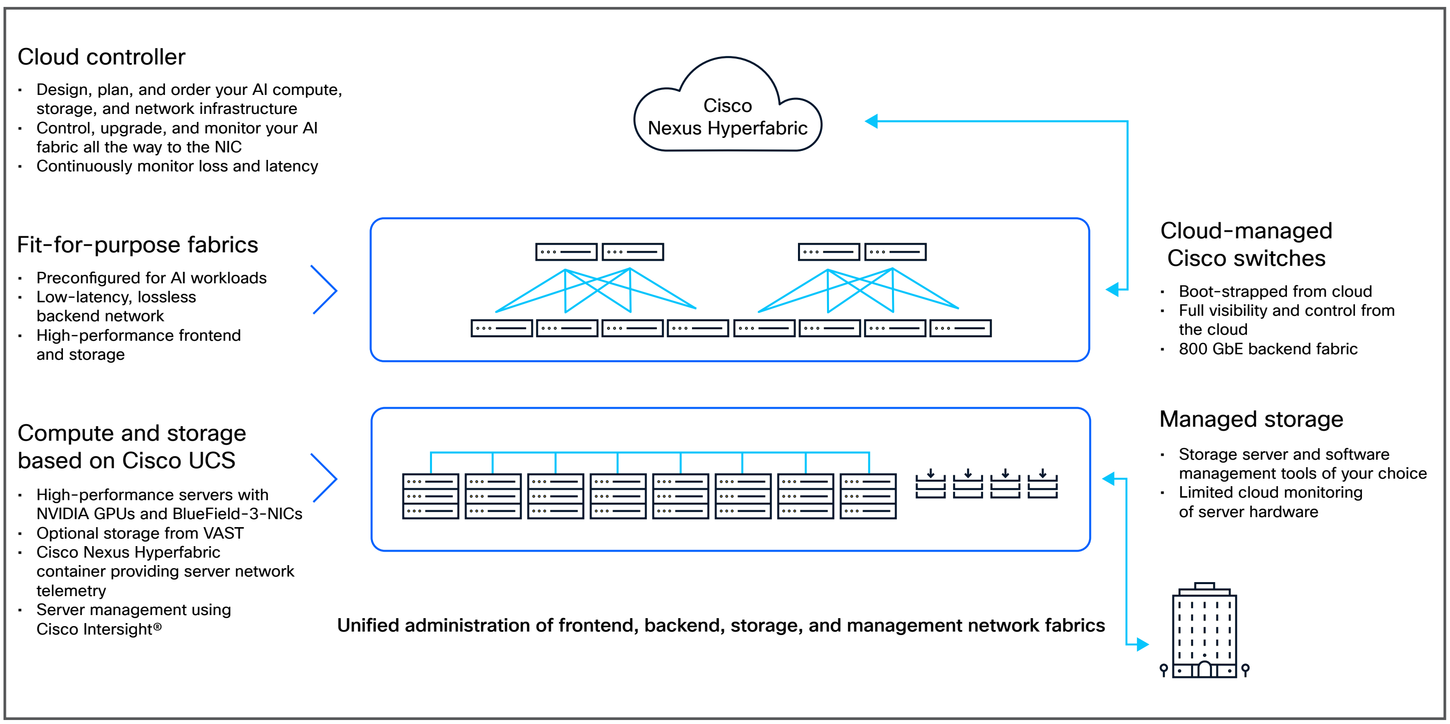

Figure 2. How Cisco Nexus Hyperfabric is built

Architecture and cloud management

Cisco Nexus Hyperfabric cloud-managed full-stack AI infrastructure simplifies the entire lifecycle management process, including design, ordering, deployment, management, and scaling of AI compute, GPUs, storage, and network infrastructure. This option features the Nexus Hyperfabric cloud controller that enables comprehensive design, planning, ordering, operation, upgrade, and monitoring of the AI fabric down to the NIC level, with continuous monitoring of loss and latency. Switches, GPU clusters, and storage servers are bootstrapped and fully visible and controlled from the cloud, ensuring seamless management.

The infrastructure includes fit-for-purpose fabrics that are preconfigured specifically for AI workloads, providing a low-latency, lossless backend network alongside high-performance frontend and storage fabrics. Software and monitoring capabilities are integrated through a Nexus Hyperfabric container that delivers server network telemetry, unified administration of frontend, backend, storage, and management fabrics, as well as assertion-based fabric monitoring and real-time health checks.

Scalability and flexibility are key features of this architecture, supporting independent scaling of network, compute, and storage resources. Its modular design allows customers to bring their own storage or opt for fully integrated turnkey options, enabling tailored deployments that meet diverse AI workload requirements. This comprehensive approach facilitates fast, reliable AI cluster deployment with a replicable model that eases AI cluster management and future scalability.

Deployment experience

The guided installation process for Cisco Nexus Hyperfabric's full stack AI infrastructure option simplifies AI cluster deployments through real-time verification, cloud-managed lifecycle, continuous software updates, and a replicable deployment model. On day 0, users access the centralized cloud-managed controller portal at hyperfabric.cisco.com to plan and design the infrastructure using the designer tool, finalize the bill of materials including optics and cables, and begin installation planning based on the cabling plan. On day 1, upon equipment arrival, Cisco 6000 Series Switches or the Cisco N9100 Series Switches, Cisco UCS servers with GPUs, and optional Cisco UCS servers with third-party storage software are registered and connected to the Nexus Hyperfabric cloud controller. The mobile-friendly portal provides step-by-step deployment guidance and real-time topology validation to ensure that the physical fabric matches the design. From day 2 onward, lifecycle management is handled remotely through the cloud controller, enabling monitoring, upgrades, and collaboration. This process automates provisioning, guides on-site tasks, and allows remote management to be accessible even to non-expert staff, thereby streamlining deployment and operational management of AI infrastructure.

Licensing

A subscription entitlement is needed for every Cisco switch that is deployed and used. Subscription entitlements may be initially purchased for three, five, or seven years and may be renewed. The subscription-feature entitlement tiers are based on fabric use cases. The “Premier” package applies to Cisco Nexus Hyperfabric’s full stack AI infrastructure option. All the switches in a fabric must use the same entitlement tier; however, an organization may concurrently manage multiple fabrics that use different entitlement tiers. For the non-full-stack AI option view the Cisco Nexus Hyperfabric data sheet.

Table 1. Cisco Nexus Hyperfabric Essentials and Premier feature tiers

Feature	Premier (Cisco Nexus Hyperfabric’s full stack AI infrastructure option only)
Cisco support 8x5xNBD	Yes
Cloud controller	Yes
Designer (no purchase required)	Yes
Cloud-driven software upgrades	Yes
BOM generation	Yes
On-site deployment assistance	Yes
Plug-and-play deployment	Yes
Spine-and-leaf topologies	Yes
Mesh (spineless) topologies	Yes
EVPN/VXLAN underlay (opaque)	Yes
Static and BGP routing	Yes
MLAG	Yes

Feature	Premier (Cisco Nexus Hyperfabric's full stack AI infrastructure option only)
Supports RDMA over Converged Ethernet v2	Yes
Real-time cloud-accessed telemetry	Yes
IPv4 and IPv6	Yes
Assertion-based monitoring	Yes
Survivable data and local management plane	Yes
Hardware-based attestation and security	Yes
API for headless provisioning and monitoring	Yes
AI use-case support	Yes
Aligned to NVIDIA Enterprise Reference Architecture	Yes
NVIDIA Adaptive Routing on backend switches	Yes
Deploy AI-validated blueprints built into workflow	Yes
Automatically provisions lossless backend AI and storage networks	Yes
Internal performance monitoring between switches and servers	Yes
Options for AI servers	Bundled Cisco UCS

Product sustainability

Information about Cisco's Environmental, Social, and Governance (ESG) initiatives and performance is provided in Cisco's CSR and sustainability reporting.

Table 2. Cisco environmental sustainability information

Sustainability topic	Reference
General: Information on product-material-content laws and regulations	Materials
Information on electronic waste laws and regulations, including products, batteries, and packaging	WEEE Compliance
Information on product takeback and reuse program	Cisco Takeback and Reuse Program
Sustainability inquiries	Contact: csr_inquiries@cisco.com
Material: Product packaging weight and materials	Contact: environment@cisco.com

Cisco and partner services

Cisco and partner services offer a wide range of services to help accelerate your success in connecting Cisco 6000 Series Switches and Cisco N9000 Series Switches to the Cisco Nexus Hyperfabric's full stack AI infrastructure option cloud controller. Our innovative service offerings are delivered through a unique combination of people, processes, tools, and partners and are focused on helping you increase operational efficiency and improve your network control. Cisco Nexus Hyperfabric's full stack AI infrastructure option provides proactive support with the Cisco SMARTnet® service to help you resolve mission-critical problems with direct access at any time to Cisco network experts and award-winning resources. Spanning the entire network lifecycle, our service offerings help increase investment protection, optimize network operations, support migration operations, and strengthen your IT expertise.

For more information, please visit www.cisco.com/go/services.

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For more information

Try before you buy

Anyone with a Cisco ID or CCO ID may log into the Cisco Nexus Hyperfabric's full stack AI infrastructure option cloud controller at hyperfabric.cisco.com to request an organization identifier; they can then begin building network fabric blueprints for free.

For more information, visit: cisco.com/go/nexus-hyperfabric-ai.

Some products and features described are in development and offered on a when-and-if-available basis. Cisco reserves the right to change delivery timelines and will have no liability for any delays or failures to deliver.

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